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Hermetic Builds with Bazel

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TL;DR: What?

```
$ BUILD_CONFIG=common/build.config.gki.aarch64 build/build.sh
```

⬇️

```
$ bazel build //common:kernel_aarch64
```

https://source.android.com/docs/setup/build/building-kernels#building-with-bazel

Android Kernel Builds

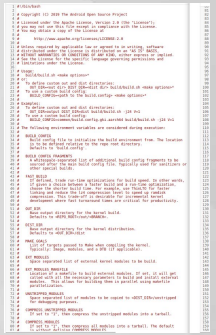
"some document" -> "build.sh" -> Bazel (Kleaf)

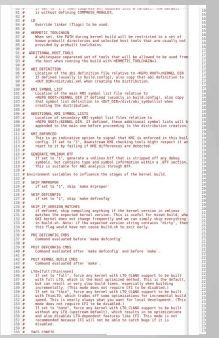
Android Kernel Builds

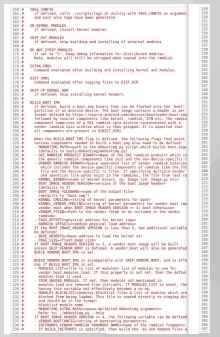
- build.sh
  - build.config configuration
  - build
  - package
  - module builds
  - mixed builds
  - ...

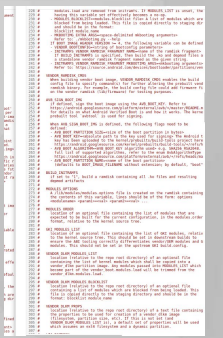
Android Kernel Builds

- build.sh
  - lots of control via env variables
  - ever-growing shell script collection
  - difficult to maintain
  - hermeticity / reproducibility as hack
  - limited parallelism









Bazel

- Scalable
- Built-in sandboxing for hermetic builds
- Fast incremental builds
- Build dependency analysis
- Parallelism
- Remote Build Execution (RBE)
- Future Android Platform build system

https://bazel.build

Kernel Builds with Bazel (Kleaf)

- Consistent environment setup
- Hermetic toolchain (tools, compilers, etc.) enforced
- Wraps make defconfig and make steps
- Kbuild remains authoritative build system

Kernel Builds with Bazel

```
load("//build/kernel/kleaf:kernel.bzl",
     "kernel_build", "kernel_module")

kernel_build(
    name = "kernel",
    outs = ["vmlinux"],
    build_config = "common/build.config.gki.aarch64",
    srcs = glob(["**"]),
)

kernel_module(
    name = "nfc",
    srcs = glob(["**"]),
    outs = ["nfc.ko",],
    kernel_build = "//common:kernel",
)
```

BUILD.bazel

- uses build.config (during migration)
- defines GKI kernel build
- kernel module against GKI

- consistent hermetic toolchain (enforced)
- sandboxed build

https://android.googlesource.com/kernel/build/+/refs/heads/master/kleaf/docs/kleaf.md

Kernel Builds with Bazel

```
kernel_build(name = "kernel", ...)

kernel_module(name = "nfc", ...)
```

```
$ bazel build nfc
```

- Builds "just enough" kernel
- Ensures limited visibility
  - kernel headers
  - scripts/
  - no other source files (.c)
- Enforces compatible toolchain
- Requires Kbuild makefiles (Kbuild, Makefile, Kconfig)

https://android.googlesource.com/kernel/build/+/refs/heads/master/kleaf/docs/kleaf.md

Driver Development Kit (DDK)

```
ddk_module(
    name = "base",
    srcs = [
        "base.c",
        "impl_base.h",  // available for module local compilation
    ],
    hdrs = ["base.h",]  // exported to other modules
    kernel_build = "//common:kernel_aarch64",
)


ddk_module(
    name = "nfc",
    srcs = [
        "nfc.c",            // can #include base.h and
        "nfc.h",            // use symbols exported from base
    ],
    kernel_build = "//common:kernel_aarch64",
    deps = [":base",],
)
```

- **Generates** Kbuild makefiles (Kbuild, Makefile, Kconfig)
- Ensures correct build dependencies
- Restricts source file visibility to declared inputs

DRAFT API

https://android.googlesource.com/kernel/build/+/refs/heads/master/kleaf/docs/kleaf.md

Useful Build Flags

`--lto={default,none,thin,full}`	LTO mode
`--use_prebuilt_gki=8728676`	Download prebuilt GKI from ci.android.com
`--config=local`	Local make cache (breaks sandbox)
`--config=fast`	--config=local and --lto=thin
`--config=release`	SCM versions, used on ci.android.com
`--kasan`	Enable KAsan
`--debug_print_scripts`	Print assembled scripts that get executed

https://android.googlesource.com/kernel/build/+/refs/heads/master/kleaf/docs/kleaf.md

Migrating to Kleaf

- build.sh deprecated
- automatic command line migration
- automatic build.config migration

Questions?